Es $r_1: \begin{cases} x+2z=0 \\ y-3z=0 \end{cases}$ $r_2: \begin{cases} x+y-1=0 \\ z=0 \end{cases}$

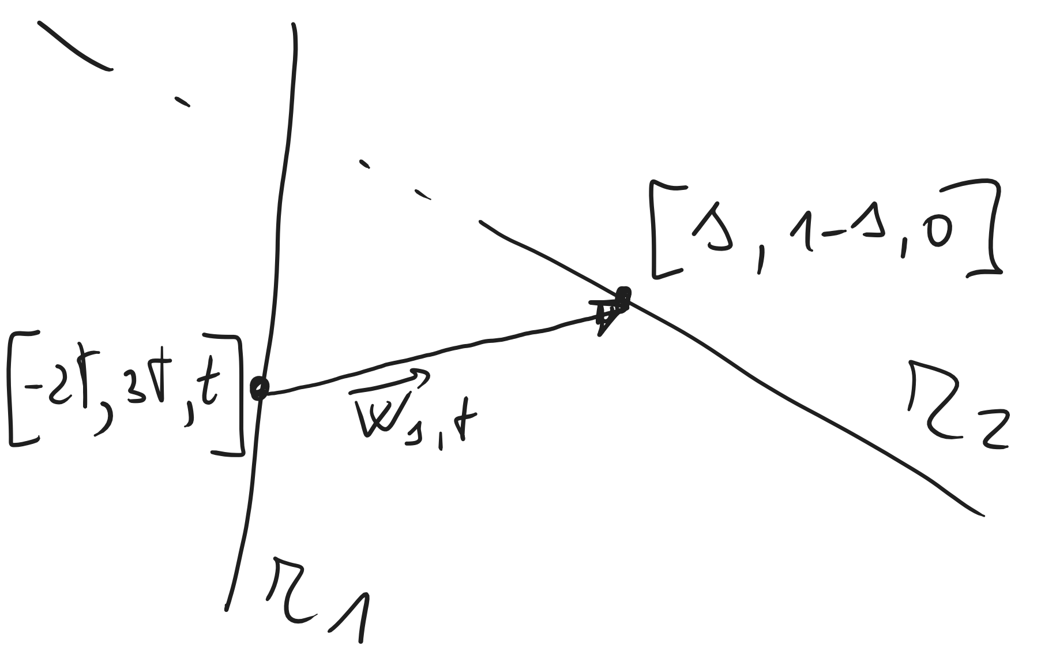
Verificare che r_1 e r_2 sono sghembe e determinare la loro comune perpendicolare r, precisando le coordinate dei punti di intersezione $Q_1 = r \cap r_1$, $Q_2 = r \cap r_2$

$r_1: \begin{cases} x=-2t \\ y=3t \\ z=t \end{cases}$ $r_1 \parallel \vec{w_1} = [-2,3,1]$ $r_2: \begin{cases} x=s \\ y=1-s \\ z=0 \end{cases}$ $r_2 \parallel \vec{w_2}$, $\vec{w_2} = [1,-1,0]$

$\vec{w_1}$ e $\vec{w_2}$ NON sono paralleli, quindi r_1 e r_2 NON sono parallele

Vediamo se r_1 e r_2 sono incidenti: $\exists\, s,t \in \mathbb{R}$ tali che

$$\begin{cases} -2t=s \\ 3t=1-s \\ t=0 \end{cases} ? \qquad \begin{cases} 0=s \\ 0=1 \\ t=0 \end{cases} \text{IMPOSSIBILE} \Rightarrow r_1, r_2 \text{ sghembe}$$



$\vec{w}_{s,t} = [s+2t,\ 1-s-3t,\ -t]$

Devo imporre $\vec{w}_{s,t} \perp \vec{w_1}$, e $\vec{w}_{s,t} \perp \vec{w_2}$, cioè $ⓧ \begin{cases} \vec{w_1} \cdot \vec{w}_{s,t} = 0 \\ \vec{w_2} \cdot \vec{w}_{s,t} = 0 \end{cases}$

$$ⓧ \Leftrightarrow \begin{cases} [-2,3,1]\cdot[s+2t,\ 1-s-3t,\ -t]=0 \\ [1,-1,0]\cdot[\quad '' \quad]=0 \end{cases} \Leftrightarrow \begin{cases} -2s-4t+3-3s-9t-t=0 \\ s+2t-1+s+3t=0 \end{cases}$$

$$\Leftrightarrow \begin{cases} -5s-14t+3=0 \\ 2s+5t-1=0 \end{cases} \quad \begin{cases} 5s+14t=3 \\ 2s+5t=1 \end{cases} \quad \begin{cases} \frac{5}{2}-\frac{25}{2}t+\frac{28}{2}t=3 \\ s=\frac{1}{2}-\frac{5}{2}t \end{cases}$$

$$\Leftrightarrow \begin{cases} \frac{3}{2}t=\frac{1}{2} \\ s=\frac{1}{2}-\frac{5}{2}t \end{cases} \quad \begin{cases} t=1/3 & Q_1=\left[-\frac{2}{3},1,\frac{1}{3}\right] \\ s=-\frac{1}{3} & Q_2=\left[-\frac{1}{3},\frac{4}{3},0\right] \end{cases}$$

$r_1: [-2t,3t,t]$
$r_2: [s,1-s,0]$

$r \parallel \vec{w}_{-\frac{1}{3},\frac{1}{3}} = \left[\frac{1}{3},\frac{1}{3},-\frac{1}{3}\right] \parallel [1,1,-1]$

Conclusione: r è la retta parallela a $[1,1,-1]$ e che passa, ad esempio, per $Q_2 = \left[-\frac{1}{3},\frac{4}{3},0\right]$. Quindi

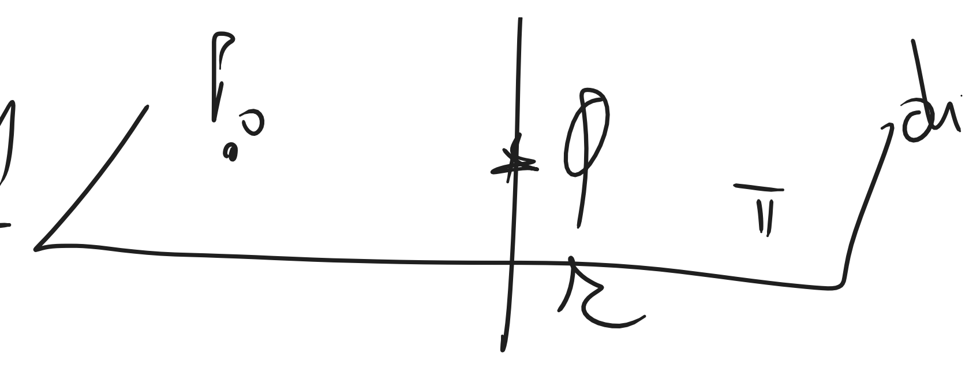
$r: \begin{cases} x=-\frac{1}{3}+t \\ y=\frac{4}{3}+t \\ z=-t \end{cases}$ $t \in \mathbb{R}$ Se voglio forma cartesiana di r "devo eliminare t"

$$r: \begin{cases} x+z+\frac{1}{3}=0 \\ y+z-\frac{4}{3}=0 \end{cases} \qquad \boxed{r: \begin{cases} 3x+3z+1=0 \\ 3y+3z-4=0 \end{cases}}$$

verificare che $Q_1 = \left[-\frac{2}{3},1,\frac{1}{3}\right]$ e $Q_2 = \left[-\frac{1}{3},\frac{4}{3},0\right]$ soddisfano

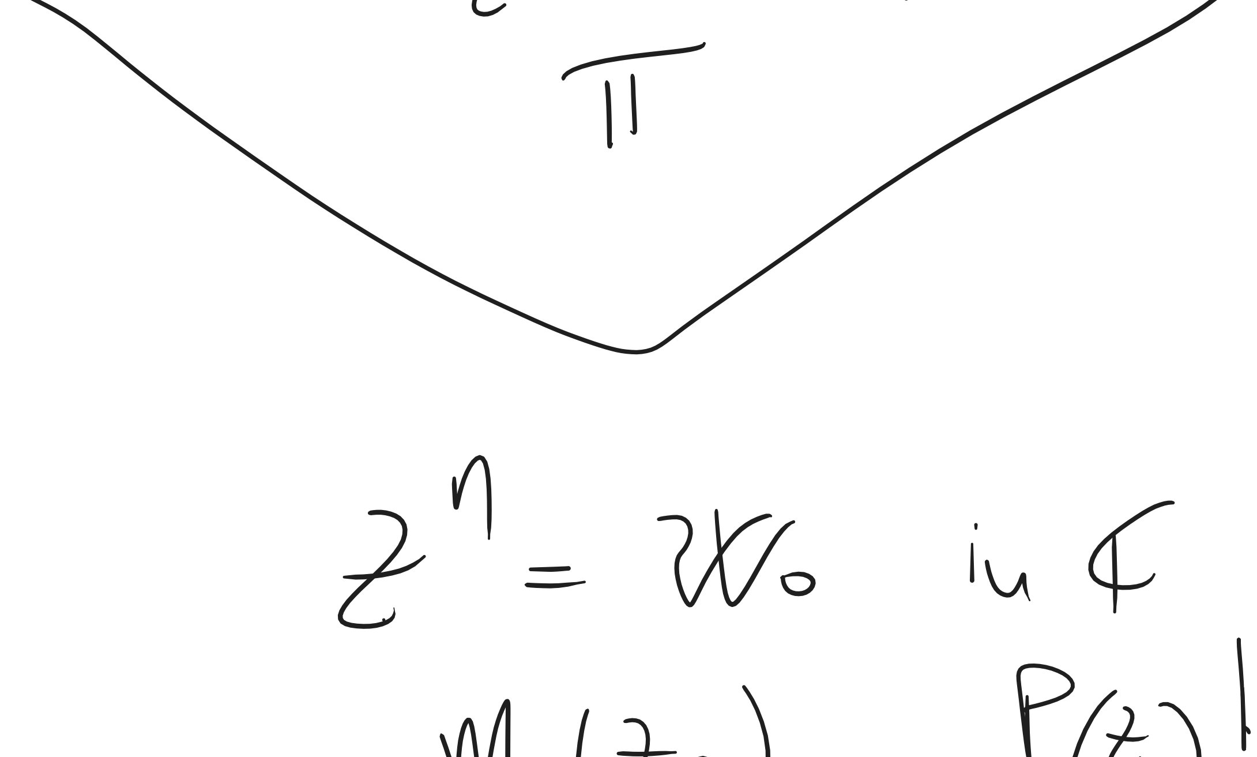
O.K.

Es1



$\text{dist}(P_0, r) = \text{dist}(P_0, Q)$

Es2



Determinare piano π che contiene r e P_0 (π appartiene al fascio di piani generato da r)

$z^n = w_0$ in $\mathbb{C}$

$m_a(z_0)$ $P(z) \mid P'(z)$